

is switched on. Once switched on, it displays the radar dial and waits for signals from the transmitter.

Next, the transmitter is switched on. The Arduino Nano sends a PWM signal to the servo motor, moving it to the 0° position. It is now ready to start scanning. The Arduino Nano then begins moving the servo motor from 0° to 180° in 1° steps. It linearly increases the width of the PWM pulse applied to the servo motor on pin D11.

Simultaneously, the Arduino Nano performs the following two tasks: (1) It transmits a scanning

sync pulse through the HC-12 to the receiver each time the motor angle is increased, and (2) It applies a trigger pulse to the HC-SR04 sensor to start its operation. When the receiver's HC-12 receives the scanning sync pulse, it sends it to the Arduino, which then starts displaying the scanning-in-progress animation on the TFT LCD radar monitor. As the motor angle increases on the transmitter side, the same scan angle is displayed on the radar monitor.

This scanning continues from 0° to 180° and back from 180° to 0° on the transmitter side. Simultaneously,

on the receiver side, the radar monitor displays this scanning process on the radar dial. This process runs continuously until no object is detected.

When the HC-SR04 sensor detects an object, it measures its distance and provides a corresponding PWM output to the Arduino. The Arduino measures the width of this pulse and calculates the object's distance. It also reads the angle of the servo motor. The Arduino then serially sends both the distance and angle of the object to the HC-12 module, which transmits the data through a 434MHz carrier to the receiver.

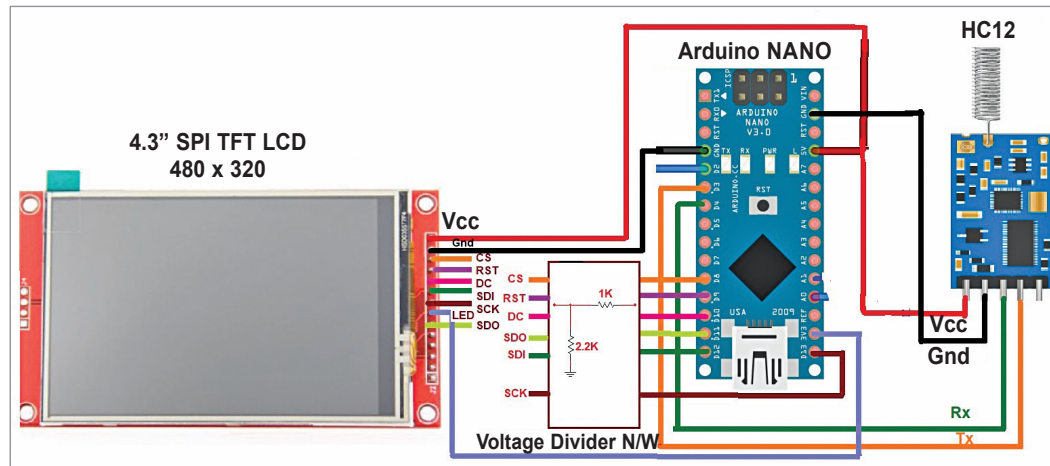


Fig. 7: Wiring diagram of receiver

On the receiver side, the HC-12 module receives the distance and angle data and sends it serially to the Arduino. The Arduino displays both pieces of data on the TFT radar monitor and indicates red line marks on the radar dial at the given angle during scanning.

Thus, the radar transmitter continu-

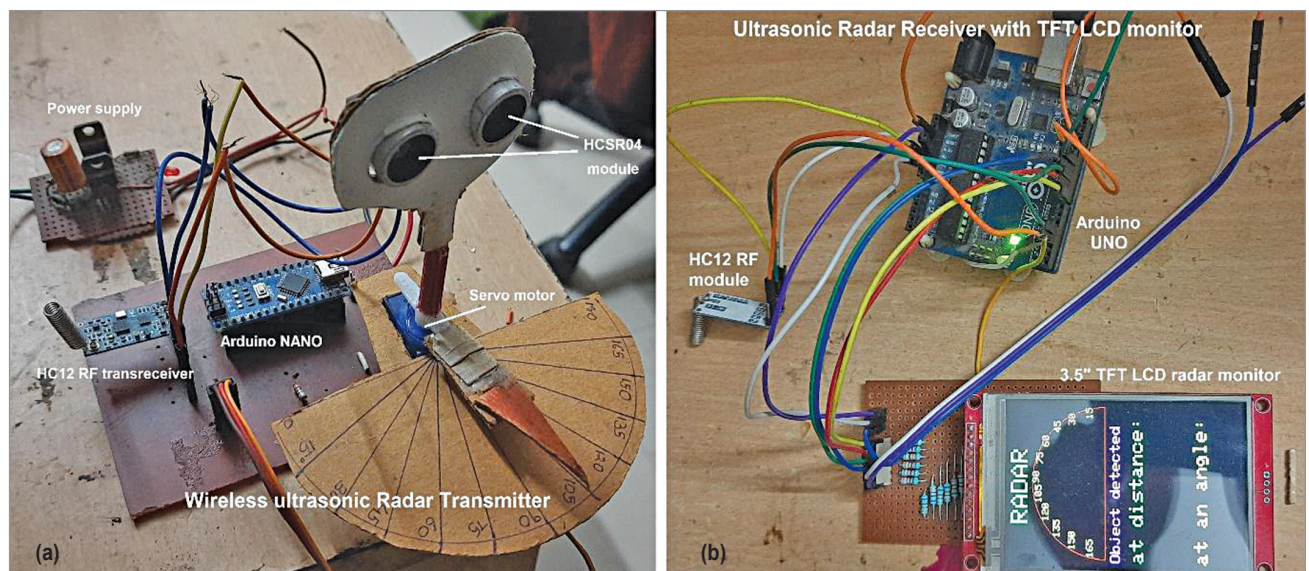


Fig. 8: Author's prototype: (a) transmitter, and (b) receiver